

CAPSTONE TWO FINAL REPORT

Loan Default Risk Prediction Using Machine Learning

1. Title

Loan Default Risk Prediction Using Machine Learning Classification Models

2. Executive Summary

In this project, I worked on predicting whether a loan applicant is likely to default using machine learning models. The main idea was to help financial institutions better understand risk so they can make more informed lending decisions.

The dataset included borrower information such as financial history, demographic details, and credit-related features. I went through the full data science process, starting with cleaning the data, exploring patterns through visualizations, preparing the features, and then building predictive models.

I built and compared three different logistic regression-based models: a baseline model, a model with an adjusted decision threshold, and a class-weighted model to handle imbalance in the data. I evaluated each model using accuracy, precision, recall, and F1-score. Since this is a risk problem, I focused more on recall, because missing a potential default is more costly than incorrectly flagging a safe borrower.

After comparing the results, I selected the model that provided the best balance between catching risky applicants and maintaining reasonable overall performance. Overall, the results showed that machine learning can add real value in credit risk assessment by identifying patterns that are not always obvious through traditional methods.

3. Problem Statement

The main problem I tried to solve in this project is predicting whether a loan applicant will default or not. This is important because lenders want to reduce financial losses while still approving good customers.

Traditional credit scoring methods do not always capture complex patterns in borrower behavior. Because of that, some risky applicants may get approved, which leads to losses for financial institutions.

The goal of this project is to build a model that can help identify high-risk borrowers more accurately. In particular, I focused on reducing false negatives, since missing a default case is more harmful than incorrectly flagging a low-risk applicant.

4. Data Description

For this project, I used a loan-related dataset that contains information about past loan applicants and whether they defaulted on their loans. The dataset includes a mix of financial, demographic, and credit-related features that help describe each borrower's profile.

The target variable in this dataset indicates whether a loan was defaulted or not, which makes this a supervised binary classification problem.

The dataset contains multiple features such as income level, credit history, loan amount, and other applicant details. These variables were used to understand patterns that might influence loan repayment behavior.

Before modeling, I reviewed the structure of the dataset to understand feature types, missing values, and overall data quality. This helped guide the cleaning and preprocessing steps later in the project.

5. Data Cleaning and Preparation

Once I explored the dataset, I performed a series of cleaning steps to make sure the data was ready for modeling.

First, I checked for missing values and handled them appropriately depending on the feature type. For numerical features, missing values were either filled using statistical measures or removed if necessary. For categorical variables, missing values were handled by using the most frequent category or an appropriate placeholder.

Next, I converted categorical variables into numerical format using one-hot encoding (dummy variables), since machine learning models require numeric input.

After that, I scaled the numerical features so that all variables were on a similar range. This step was important because logistic regression is sensitive to differences in feature magnitude.

Finally, I split the dataset into training and testing sets so that model performance could be evaluated on unseen data.

6. Exploratory Data Analysis (EDA)

Before building the models, I explored the dataset to better understand relationships between features and the target variable.

I created a few key visualizations to identify patterns and potential insights:

- A class distribution plot to check whether the dataset was balanced or imbalanced
- A correlation heatmap to understand relationships between numerical features
- Distribution plots for key variables such as income and credit-related attributes

From this analysis, I observed that some features had stronger relationships with loan default than others. I also noticed that the dataset had some level of class imbalance, which influenced my decision to include a class-weighted model later.

Overall, the EDA helped me better understand which features might be useful for predicting loan default risk.

7. Feature Engineering

To improve model performance, I performed feature engineering on the dataset.

Categorical variables were converted into dummy variables using one-hot encoding. This allowed the model to properly interpret non-numeric data.

I also standardized the numerical features so that variables with larger scales did not dominate the model. This step was especially important for logistic regression.

After preparing the features, I reviewed the dataset again to ensure everything was in a clean numerical format suitable for modeling.

8. Modeling Approach

After preparing the data, I moved on to building machine learning models to predict loan default risk. Since this is a binary classification problem, I chose Logistic Regression as the main modeling technique because it is interpretable and works well for baseline risk prediction problems.

I built and compared three different variations of Logistic Regression to see how performance changes under different settings:

1. Baseline Logistic Regression

This was the first model I built using default settings. It served as a reference point to understand how the model performs without any adjustments.

2. Logistic Regression with Adjusted Threshold

In this version, I changed the decision threshold instead of using the default 0.5 cutoff. This was done to see if the model could better capture high-risk borrowers by being more sensitive to predicting defaults.

3. Logistic Regression with Class Weights

Since the dataset showed some level of class imbalance, I used class weighting to give more importance to the minority class (loan defaults). This helped the model pay more attention to risky cases instead of being biased toward the majority class.

All models were trained using the training dataset and evaluated on the test dataset to ensure fair performance comparison on unseen data.

9. Model Evaluation

To evaluate the models, I used four key metrics:

- Accuracy
- Precision
- Recall
- F1-score

These metrics were selected to provide a balanced evaluation of model performance. However, in this problem, recall is especially important because the cost of missing a loan default (false negative) is significantly higher than incorrectly flagging a safe applicant. In a financial context, it is more important to correctly identify risky borrowers than to simply maximize overall accuracy.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.99	0.99	0.96	0.97
Random Forest Classifier	0.95	0.91	0.84	0.87
Logistic Regression (Tuned / Final)	0.99	0.99	0.96	0.97

All models performed well, but Logistic Regression showed the strongest and most consistent performance, especially in identifying loan defaults.

The Random Forest model also performed strongly and demonstrated good general predictive ability. However, it had lower recall for the default class compared to Logistic Regression.

This may indicate that the dataset has strong linear separability, which allows Logistic Regression to perform extremely well compared to more complex models like Random Forest.

Overall, recall and F1-score were prioritized when comparing models, since correctly identifying loan defaults is more important than overall accuracy in this business context.

10. Final Model Selection

Based on the evaluation results, the final model selected for this project is the Logistic Regression model.

Although Random Forest performed well in capturing general patterns in the data, Logistic Regression was chosen as the final model because it achieved the highest and most consistent recall for the default class, which is the most important metric for this problem.

In a loan default prediction scenario, the primary goal is to correctly identify risky borrowers in order to minimize financial loss. Logistic Regression performed better in this aspect while also maintaining strong overall accuracy and stability.

Therefore, the Logistic Regression model was selected as the final model because it consistently achieved the highest recall for the default class while maintaining strong overall performance. This makes it the most suitable model for identifying loan default risk in a real-world lending scenario.

11. Results and Business Interpretation

After evaluating the models, the results showed that all three Logistic Regression variations performed reasonably well, but they behaved differently depending on how they were tuned.

The baseline model provided a good starting point but was not optimized for identifying high-risk borrowers. The threshold-adjusted model improved sensitivity by capturing more potential default cases, but it also increased false positives. The class-weighted model provided the most balanced performance by improving the model's ability to detect minority class cases (defaults) without heavily sacrificing overall accuracy.

From a business perspective, the most important outcome is the model's ability to correctly identify applicants who are likely to default. Even if some low-risk applicants are incorrectly flagged, it is generally safer for a financial institution than approving high-risk borrowers.

Overall, the results show that adjusting the model to prioritize recall leads to better risk control in lending decisions.

12. Recommendations

Based on the analysis and model results, I would recommend the following actions:

- 1. Use the model as a pre-screening tool**

The model can be used to flag high-risk loan applicants before final approval. This helps reduce the chances of approving borrowers who are likely to default.

- 2. Prioritize recall in decision-making**

Since missing a default case is more costly than a false alarm, the institution should prioritize models and thresholds that maximize recall, even if it slightly reduces accuracy.

- 3. Combine model predictions with traditional credit scoring**

Instead of replacing existing systems, the model should be used alongside traditional credit evaluation methods to improve overall decision quality.

4. Periodically retrain the model

Borrower behavior and economic conditions change over time, so the model should be retrained regularly with updated data to maintain performance.

13. Future Work

Although the current model provides useful insights, there are several ways it could be improved in future iterations:

- Testing more advanced models such as Random Forest, XGBoost, or Gradient Boosting
- Performing deeper feature engineering to capture more behavioral patterns
- Handling class imbalance using additional techniques such as SMOTE
- Incorporating external economic indicators to improve prediction accuracy

These improvements could potentially lead to even stronger predictive performance.

14. Conclusion

This project demonstrated how machine learning can be applied to predict loan default risk using historical borrower data. By comparing multiple Logistic Regression models, I was able to identify a model that better captures high-risk applicants.

The final model provides meaningful value for financial decision-making by helping reduce lending risk and improving the quality of loan approvals. While the model is not perfect, it offers a strong foundation that can be improved further with additional data and more advanced techniques.

Overall, this project shows that data-driven approaches can significantly enhance traditional credit risk assessment processes.